Lesson 02: Functions and Data Quick Reference

Activity Finder

Functions, parameters and return

Arguments are supplied when a function is called. Parameters receive them. return sends a result to the caller.

```
function label(activity) {
  return activity.title;
}
```

Scope

Variables declared inside a function cannot be used by name outside that function. A function may read enclosing variables, but parameters make required inputs visible.

Objects and arrays

One object groups related properties; an array can contain several objects.

```
const activities = [
  { title: "Chess Club", category: "Games", location: "Studio 2", isFull: false },
  { title: "Coding Workshop", category: "Technology", location: "Lab 4", isFull: true },
  { title: "Cycling Group", category: "Sport", location: "Sports Hall", isFull: false }
];
```

Callbacks and arrow functions

filter() and map() call a callback once per activity. Braced arrow bodies require an explicit return.

```
(activity) => { return activity.title; }
(activity) => activity.title
```

filter() selects

The callback returns a truthy value for items to keep. filter() returns a new array containing selected elements.

```
const matches = activities.filter((activity) => {
  return activity.title.toLowerCase().includes(query);
});
```

map() transforms

The callback return becomes one element in the new array. Missing return often produces undefined values.

```
const titles = activities.map((activity) => {
  return activity.title;
});
```

const with arrays and objects

const prevents reassignment of the variable. It does not freeze array contents or object properties.

```
const activity = { isFull: false };
activity.isFull = true;  // allowed
// activity = {};        // reassignment error
```

Common-error checklist

Check property names, callback return values, case normalisation, expected output and whether code changed a supplied object unexpectedly.

Useful references

MDN Functions, Array.filter(), Array.map() and const. Use only the section relevant to your task.